Future of Algorithmic Organization: Large Scale Analysis of Decentralized Autonomous Organizations (DAOs)

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Decentralized Autonomous Organizations (DAOs) resemble early online communities, particularly those centered around open-source projects, and present a potential empirical framework for complex social-computing systems by encoding governance rules within "smart contracts" on the blockchain. A key function of a DAO is collective decision-making, typically carried out through a series of proposals where members vote on organizational events using governance tokens, signifying relative influence within the DAO. In just a few years, the deployment of DAOs surged with a total treasury of $24.5 billion and 11.1M governance token holders collectively managing decisions across over 13,000 DAOs as of 2024 [1]. In this study, we examine the operational dynamics of 100 DAOs, like pleasrdao, lexdao, lootdao, optimism collective, uniswap, etc. With large-scale empirical analysis of a diverse set of DAO categories and smart contracts and by leveraging on-chain (e.g., voting results) and off-chain data, we examine factors such as voting power, participation, and DAO characteristics dictating the level of decentralization, thus, the efficiency of management structures. As such, our study highlights that increased grassroots participation correlates with higher decentralization in a DAO, and lower variance in voting power within a DAO correlates with a higher level of decentralization, as consistently measured by Gini metrics. These insights closely align with key topics in political science, such as the allocation of power in decision-making and the effects of various governance models. We conclude by discussing the implications for researchers, and practitioners, emphasizing how these factors can inform the design of democratic governance systems in emerging applications that require active engagement from stakeholders in decision-making.

CCS Concepts: • **Information systems** → *Collaborative and social computing systems and tools.*

Additional Key Words and Phrases: DAO, Decentralized Governance, Organizations

[1] our count of 13,000 follows the broader definition of DAOs that includes active, dormant or treasury-less entities.

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1 Introduction

Decentralized Autonomous Organizations – closely similar to early online communities, especially those focused on open-source projects. They also draw inspiration from various models, including digital and platform cooperatives [46], multi-organizational networks like keiretsus [38], crowdfunding platforms such as Patreon, virtual economies in games like World of Warcraft and Second Life [36] and peer-produced projects like Wikipedia [79]. DAOs challenge the traditional roles of firms by providing a potential empirical testbed for exploring social choice experiments in potentially improving the current governance structure as well as improving longstanding social, organizational, and legal issues through a computational lens [5]. As such, DAOs present unique opportunities to implement innovative mechanisms in social choice designs, including, quadratic voting, and forms of liquid democracy [33, 75, 81]. DAO governance, as human-centric digital organizations, addresses key issues in social computing but is more complex than platforms like civic tech [57], blockchain governance, online communities [42]. DAOs emphasize entrepreneurial functionality with less reliance on existing institutional hierarchy, with greater emphasis on legal, economic, and organizational sophistication. They prioritize governance over traditional management tools (e.g., Slack, Salesforce, Zendesk), which do not manage rights and power digitally.

In just a few years, the deployment of DAOs surged with a total treasury of $24.5 billion and 11.1M governance token holders collectively managing decisions across over 13,000 DAOs as of 2024 [2]. Technologists argued that DAOs could automate many organizational processes and allow for more broad-based ownership and governance of the digital economy, all on the basis of a cryptographically secured blockchain [10]. DAOs transcend being mere novelties within online communities; they possess the transformative power to shift vast segments of institutional frameworks from the physical realm to the digital world, merging legal and economic principles with computer science. DAOs in their current form may or may not become the future of organizations, the increasing significance of online organizational forms in global politics and economy is undeniable. Research in DAOs presents a promising avenue for addressing complex issues related to organization, coordination, and governance, marking it as a critical area of study in the evolving landscape of digital and organizational sciences.

A key function of a DAO is collective decision-making, typically carried out through a series of proposals where members vote on organizational events using governance tokens, signifying relative influence within the DAO. The process of voting on proposals is central to the governance of DAOs. Voting raises many design decisions, such as how to allocate voting power and how voting choices should translate into outcomes [62]. These decisions relate to questions of social choice [64] and tokenomics. DAO governance relies on secure and trustworthy voting mechanisms, which differ significantly from traditional political voting. First, DAO voting is entirely digital, eliminating the need for physical polling stations or ballots. Second, it is typically pseudonymous, with voters identified by on-chain addresses instead of real-world identities. Finally, unlike most political elections that allow one vote per eligible voter, DAO voting often employs various methods that weight votes based on token holdings. DAO voting bears some similarity in this sense to corporate proxy voting. In previous works, researchers highlighted social and technical metrics such as token distribution, voter participation, voting delegation, and voting behavior patterns [21, 62]. Some tension arises from social measures such as trust and community belonging, which influence decision-making and key functions within economic coordination such as, voting power distribution and equality, and level of decentralization [7, 23, 62].

Building on existing case studies of a limited number of DAOs to identify factors influencing their success or failure, as well as stakeholder interviews [21, 62], individual protocols, specific governance mechanisms [12, 24, 70, 73], this study advances the literature by providing a large-scale, cross-category empirical analysis of realized voting power across 100 DAOs spanning social, infrastructure, investment, DeFi, and public-good domains. We aim to evaluate the significance of various governance components that enhance governance quality. We conducted a large-scale empirical analysis of 100 DAOs with a diverse set of categories and smart contracts

[2] https://deepdao.io/organizations

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by leveraging on-chain (e.g., voting results, proposal data) and off-chain data. We examine how governance mechanisms, such as, voting power, token design contribute to varying levels of decentralization across different DAOs. In particular, this paper addresses the following research questions.

RQ1: What structural and contextual factors (e.g., DAO type, governance token design, treasury size, market value) are associated with higher or lower levels of decentralization in DAOs?

RQ2: How do voting mechanisms shape the distribution of power and decentralization outcomes in DAOs?

Main Findings. Our analysis across various DAOs revealed statistically significant results between DAO properties—such as token market value, DAO type, and voting power distribution—and decentralization levels. We found social or public-good-focused DAOs (e.g., sports DAOs) showing lower Gini coefficients, indicating higher decentralization, while infrastructure and investment DAOs demonstrated greater centralization. DAOs with higher secondary market token values often had poor decentralization. We observed notable power imbalances in DAOs like 1inch, Balancer, dYdX, Fei, GMX, and Gnosis, with low voter participation and high centralization. In our analysis, we found Gini metrics to be consistent across DAOs, suggesting it can be leveraged by stakeholders to monitor decentralization trends over time. Our findings also highlight that the governance mechanisms, such as, power distribution, incentive design, and grassroots participation can inform design choices for governance in emerging areas such as AI and technology governance, where researchers and practitioners are actively exploring potential democratization among various stakeholders [1].

2 Related Work

2.1 Background of DAOs

The concept of a DAO has existed since the mid-2010s [13] when DAOs were envisioned as digital alternatives to conventional organizations, promising automation of organizational processes and broader ownership and governance in the digital economy on the basis of a cryptographically secured blockchain [10, 35]. While DAOs with such operations existed mostly in the realm of speculation [17], the reality of DAOs has drastically changed over the past few years. While some researchers initially argue that DAOs were limited to private capital allocation [13, 71], such as, Digix.io[3], Augur[4], Uniswap[5], there is a growing trend to use DAOs in high-value data, and reputational-based systems [3, 13, 48, 50, 74] and even the first DAO, *The DAO*, was originally designed as an investor-driven venture capital fund that relied on voting by investors to disburse funds to proposals submitted by contractors and vetted by curators [47]. It operated as a transparent and democratically structured virtual platform, without formal managerial roles.

Recent work has also expanded the understanding of DAO governance from multiple perspectives. For instance, voting-bloc entropy [18] introduces a novel metric that accounts for coordinated voting blocs, offering a more realistic measure of decentralization than simple token distributions, and demonstrating vulnerabilities to bribery and delegation. Complementing this, anotehr work develops a KPI-based framework to evaluate DAO governance risks, highlighting systematic weaknesses and providing empirical analysis using on-chain datasets [44]. On the design side, a recent work also synthesizes organizational and architectural patterns for DAOs, outlining both challenges and benefits for building robust Web3 systems [80]. These studies advance DAO research by linking decentralization metrics, governance security, and architectural design into a more comprehensive perspectives. We approach some concrete questions around power and participation through empirical evidence, while also considering how key components of DAO governance may be leveraged for mainstream governance contexts such as artificial intelligence oversight, citizen science, and technology governance.

[3]Digix.io is a smart-asset gold-focused coin.
[4]Augur centers around the prediction markets where financial options and insurance markets can be developed.
[5]Uniswap is a crypto exchange based on smart contracts

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While the primary objective of a DAO is to replicate traditional organizations with distributed decision-making by using blockchain technology, the precise definition of a DAO is currently contested [17]. Some argue that all cryptocurrencies can be considered DAOs, while others propose that decentralized and autonomous governance structures could be considered the original DAOs [49]. As such, the specific tasks that require coordination within a DAO and what it means to *"coordinate"* in this context of blockchain require further exploration. By empirically examining the factors that shape decentralization and the role of voting mechanisms, our study aims to bridge this gap. In doing so, our work will inform mainstream governance, offering insights into emerging models for collective decision-making in areas such as AI regulation, participatory civic engagement, and technological infrastructure management. Recent studies have begun to examine governance security, protocol risks, and voting behaviors in specific technical contexts [18, 70, 73]. For example, recent work has analyzed governance vulnerabilities, delegation [18], and protocol-level coordination dynamics [44]. We focus on an analytical lens of cross-DAO comparative perspectives. By sampling across heterogeneous DAO categories and governance infrastructures, we surface structural relationships between market context, participation, and power concentration in an ecosystem-level empirical mapping of governance patterns.

2.2 Co-ordination in Social Collaborative Systems

Governance, particularly coordination, plays a crucial role in project-based activities within online communities, open-source projects, and collective intelligence platforms such as Wikipedia [72], Slashdot [57], the Linux Kernel [42], and Zooniverse [65]. Previous research highlights the systematic benefits these communities offer compared to markets and hierarchical management structures in digitally networked environments, especially in information or cultural production [4]. The success of such communities depends on effective coordination, governance, trust, and the scalability of collaboration [32]. While not always explicitly labeled as DAOs (Decentralized Autonomous Organizations), a wealth body of literature in areas like the theory of firms [77], public choice theory [64], and platform cooperatives [54, 61] addresses themes such as decentralized governance, incentive alignment, self-management, group dynamics and paradoxes in decision making [53], themes central to the structure of DAOs. DAOs transcend being mere novelties within online communities; they possess the transformative power with technical solutions to shift vast segments of institutional frameworks, merging legal and economic principles with computer science.

Furthermore, coordination in management science is critical to ensure that resources are used efficiently and that organizations work towards the same objectives [20]. Performance metrics and feedback mechanisms [20], as well as computer-based coordination tools [22, 67], are deemed to be necessary to track progress towards organizational goals. Similarly, DAOs, as internet-native organizations, manage coordination digitally with a protocol specified in code and enforced on the blockchain. When environmental change is high, organizational systems need to adapt quickly, and this work is typically facilitated by people who focus almost exclusively on coordination as opposed to execution and that is the role of management, or, put into the language of DAOs, that is the role of community managers and delegates [9]. DAOs represent a unique combination of the concepts of Autonomous Agents (without human involvement) and Decentralized Organizations (without external influence), blurring the boundaries between the two.

Another important factor in online community is the influence of ones' voice in the community [25, 40], akin to the voting power in DAOs [2, 25, 26, 37, 43, 55, 62]. Recent studies suggest that governance tokens in DAOs can be vulnerable to bribery, affecting decision-making processes. This is comparable to both online and offline communities where influential users can manipulate recommendations. Bribery in these contexts can exploit social influence and create biased outcomes[58]. Voting paradoxes and the Gibbard-Satterthwaite theorem [6] highlight that an individual's voting power can be influenced by the structure of the voting system and the distribution of preferences, and no system is perfect at consistently representing voter preferences [60]. Although prior research has explored the voting power of DAO holders on the Ethereum blockchain [23], it has not fully

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addressed the importance of participation, which is crucial for legitimizing decisions [55]. In this study, we aim to investigate the relationship between individual influence and participation, proposing design suggestions to encourage broader adoption of DAOs.

2.3 DAOs in Technological Governance

Scientists and practitioners have been quick to acknowledge the potential power of DAOs in the context of Governance, and institutional economics, organizational imprinting [1, 63, 69]. The emergence of Decentralized Autonomous Organizations (DAOs) introduces possible solutions for various challenges related to political institutions, including classic coordination dilemmas such as preference aggregation, credible commitments, audience costs, information asymmetry, representation, and accountability [27]. This unique empirical context offers an opportunity for political scientists to examine fundamental theories concerning political institutions and develop innovative theories that can be tested within digital governance. Political institutions encompass both formal and informal rules, procedures, and organizations governing individuals' and groups' behavior within a political system [27]. These institutions represent the *rules of the game* or the constraints shaping human interaction [51]. Scholars have engaged in debates about the consequences of different institutional designs, such as the separation of powers, federalism, the strategic configuration of non-democratic institutions, and processes of institutional change. The relevance of these theories to the design of digitally-native governance institutions is a critical question. For instance, the separation of powers in DAOs impacts the prevention of excessive concentration of power, enhances transparency and accountability, or potentially leads to governmental gridlock and indecision [15]. Governance questions have puzzled societies and organizations, with the contested concept of democracy, a system contingent on the will of the people, occupying a central place in this debates [14, 34, 59]. This has gained new relevance with emerging technologies like AI, where inclusive and representative decision-making is critical throughout the development lifecycle. In theory, DAOs present innovative opportunities for collective decision-making. However, challenges persist, especially concerning technocracy, which continues to be addressed through adaptable mechanisms like quadratic funding [11] and automated decision-execution protocols.

3 Method

3.1 Selection Criteria & Method

We selected 100 DAOs, which were formed between 2019 and 2023, prioritizing diverse categories (e.g., investment, defi, social, etc), popularity, market capitalization, type of Treasury, and the nature of governance token.

DAO types. DAO types mainly refer to the primary functionality and operational focus of the organization, both at the community and smart contract levels. For instance, Sushiswap and Lido DAO are DeFi protocols, while CityDAO and LinksDAO are categorized as social DAOs.

Type of Treasury. DAO treasury is the pool of funds for the continuous growth and development of the organization. Members of the DAO rely on the specified governance mechanisms for determining the allocation of the DAO treasury funds. We referenced DAOHQ[6], DeepDAO[7], popular publicly available analytics platform of DAOs, to determine the threshold of treasury among currently available DAOs. Generally, the threshold of the treasury is divided into four main categories, including (0-$1m, $1m - $10m, $10m-$50m, $50m+). We utilized this threshold to select DAOs with diverse treasury token holdings. We distributed 0-1m as low treasury, $1-10m as medium, and $10-50m+ as high treasury.

Nature of governance tokens (monetary vs. non-monetary). We used DAOs with no secondary market value as non-monetary where those types of DAOs largely fall under social, research DAOs. More specifically,

[6]DAOHQ: https://www.daohq.co/
[7]DeepDAO: https://deepdao.io/organizations

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governance tokens that cannot be sold in the secondary market, such as the reputation-based and nontransferable dxDAO token.

Stratified Sampling. We applied a stratified sampling method to select 100 DAOs from a pool of 2391 DAOs in DeepDAO analytics. To ensure proportional representation across all factors, we divided the population (n:2391) into mutually exclusive, collectively exhaustive groups (strata) that are relevant to the goal of this study, and then we sample within each stratum. We divided the population into strata based on all four factors. For each of the factors, DAOs are subdivided into different groups where DAOs in each group are similar to one another. Every stratum is sampled and therefore represented. The four factors, we utilized are (1) DAO Types, (2) DAO Treasury Amount, (3) DAO voting module, and (4) the Nature of the governance token. In constructing the dataset, certain dimensions (e.g., DAO type, nature of governance token) required annotation. We derive this from classifications from official DAO documentation such as whitepapers, governance portals, and websites, supplemented by secondary sources including community forums and aggregator platforms (e.g., DeepDAO). There were ambiguities where one DAO can fall under several categories or non-decided categories in the data sources. We resolved through cross-checking and team discussion to ensure consistency across the sample. For each of the factors, there are subgroups where the DAOs are similar to each other. For example, for factor (1) DAO types has 10 strata DAOs such as Defi protocol, investment, DAO tool, NFTs, research, etc. For factor (2) treasury size, our population has 4 strata, including DAO with $\$0 - \$1m$, $\$1 - \$10m$, $\$10m - \$100m$ and 100m+. For factor (3) DAO voting module, there are three strata, including (a) snapshot, (b) on-chain, and (c) Others. For factor (4) Nature of DAO tokens, there are two types (a) monetary and non-monetary. Once we have defined the strata, we assign to exactly one subgroup for each factor with GraphQL queries for each characteristic. Within each stratum, we then randomly sampled DAOs proportional to the stratum's size. This ensures the original population distribution and prevents overrepresentation of any single group. The entire pipeline, including DAO selection, stratum assignment, and sampling, is available in a public GitHub repository [8]. After sampling, we compared the distributions of DAO type, treasury size, voting module, and governance token across the selected 100 DAOs against the full population of 2,391 DAOs to ensure that our sample adequately represents the diversity of the population.

3.1.1 Signaling vs. Binding Governance. Traditionally in DAOs, Snapshot is used for two fundamentally different ways, including (1) as a non-binding signaling mechanism (e.g., opinion polling or temperature checks), and (2) as a de facto binding governance mechanism where vote outcomes directly determine execution, either socially or through subsequent automated processes. In our case, we classify DAO as binding Snapshot governance if Snapshot vote outcomes are treated as final decisions. In contrast, we classify Snapshot as signaling when votes are explicitly described by the DAO as advisory, followed by a separate binding decision, typically via on-chain governance or multisig execution. We followed a manual review of DAO governance documentation, including governance portals, forum posts, proposal templates, and execution workflows. We annotated the dataset *"False"* when the Snapshot voting data is not treated as a direct measure of power execution rather as preference signaling. For instance, DAOs that use Snapshot purely for non-binding signaling or temperature checks, followed by binding execution through on-chain governance or multisig processes, are recorded as $VotingPlatform = Snapshot$ but $Snapshot = FALSE$. Thus, the binary indicators denote binding decision mechanisms, not merely the presence of a voting interface. In our dataset, DAOs such as MatrixDAO and LexDAO use Snapshot to surface community preferences, while final decisions are executed through on-chain or off-chain administrative mechanisms; therefore, these DAOs are marked as $Snapshot = FALSE$ despite listing Snapshot as their voting platform.

[8]https://github.com/TanusreeSharma/DAO-Large-Scale

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3.2 Data Collection

To address the research question, we identify and collect the necessary data by automating the process for voting data. We derive voting power distributions directly from governance platforms (e.g., Snapshot, on-chain modules) at the proposal level. For each proposal, the effective voting weight of every participant was recorded as reported by the platform, which already accounts for mechanism-specific rules (e.g., token-weighted, quadratic, etc). This is to capture the realized influence of participants rather than theoretical token balances. Because membership and voting power change over time, we aggregated distributions across all proposals for each DAO during the study period to obtain comparable measures of decentralization.

Our objective is to test the influence of various factors on the level of polarization in DAOs, similarities/differences in their voting behavior, and the degree of decentralization over time. These factors include the token holdings of DAO holders, tokens with higher secondary market value, the voting choices (for or against) made by members, voting power of DAO members, and participation. To accomplish this, we require a dataset that encompasses the voting data of DAO members, including attributes such as DAO holders' addresses, delegated voting power during voting for different proposals, timestamps, proposal IDs, and voting choices for different proposals. The majority of the DAOs leveraged the snapshot platform for voting, while some utilized on-chain voting or other platforms such as the Aragon app or manual community consensus. In our analysis, voting modules (Snapshot, on-chain governance, and other/manual methods) were identified for all DAOs and were used exclusively as stratification criteria in our sampling process, to ensure our final dataset reflected the diversity of governance infrastructures across the DAO ecosystem. We did not include voting module type as an explanatory variable in regression analyses, since our focus was on voting power and participation-related predictors of decentralization. Accordingly, we developed four types of voting data crawlers tailored to the different voting methods employed by the DAOs. For DAOs that operate completely on-chain (e.g., makerDAO), we utilized libraries like Web3.js or Ethers.js, along with the Python programming language, to query relevant contract methods and retrieve information about proposals, votes, voter details, timestamps, and other pertinent data points. For platforms like Snapshot, we utilized publicly available GraphQL APIs to crawl the voting data. As a result, our dataset contains 100 DAOs, 606563 voter addresses, and 10939 proposals.

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3.3 Data Analysis

Drawing from the related work (section 2) we formulate hypotheses.

Factors Impacting Level of Decentralization. We conduct statistical analysis considering the following factors and decentralization levels using Gini, Nakamoto, and entropy metrics where predictors are subsequently, Marketcap, Treasury, Token secondary market, and DAO Type.

Hypotheses

- **H1: Governance Token**: DAOs with tradable governance tokens exhibit higher concentration of power compared to DAOs without tradable tokens. (linked to RQ1)
- **H4: Treasury**: DAOs with larger treasuries exhibit greater centralization due to increased barriers to participation. (linked to RQ1)
- **H2: DAO Type**: Different DAO types are more likely to influence the level of decentralization (linked to RQ1)
- **H3: Market Cap**: DAOs with a high market cap are more likely to have a lower decentralization (linked to RQ1)
- **H5: Participation Rate**: An upward trend in grassroots participation is more likely to increase the trend in decentralization in a DAO (linked to RQ2)
- **H6: Voting Power**: A lower variance in voting power within a DAO is more likely to increase the level of decentralization in a DAO. (linked to RQ2)

Evaluation of Decentralization Levels with Voting Power. We conducted a correlation analysis between voters' token weight and a participation rate, which is defined as the number of proposals that a voter participated in. We aim to determine whether the participation rate, therefore the voting power, is biased toward rich token holders. Here, the positive correlation between the participation rate and token weight implies poor decentralization due to the more active participation of only richer voters than their counterparts. Next, we evaluated the decentralization level of 100 DAOs in terms of voting power. To do this, we used three metrics: Entropy, the Gini, and Nakamoto coefficient. Entropy can be generally used to assess randomness and degrees of freedom for a given system. Mathematically, the metric for given a series of numbers $x(= x_i)$ is defined as follows: $Entropy(x)=\sum_i log_2\left(\frac{x_i}{\sum_i x_i}\right) * \frac{x_i}{\sum_i x_i}$. If the value of entropy is high, it implies a DAO has a high degree of randomness and freedom, which is related to a high level of decentralization. The Gini coefficient used to measure wealth inequality traditionally can be applied to assess the voting power inequality among voters in our analysis. The metric for given a series of numbers x is defined as follows: $Gini(x)=\frac{\sum_{i,j}|x_i-x_j|}{2|x|\sum_i x_i}$, where $|x|$ indicates the length of the number series. The minimum value of the Gini coefficient as 0 would represent the full equality among voters, while the maximum value of 1 would represent the full power bias towards one voter. We calculated Entropy and Gini based on the token weight of voters and their voting scores, which, in this study, we define as *"token weight×the number of proposals that a voter participated in"* to reflect not only the token weight but also a participation rate. The participation-weighted voting power is meaningful because the influence of "active" participants is what ultimately shapes DAO governance. For instance, even if a voter holds substantial voting power, that power has no effect on proposals in which they do not participate. Lastly, the Nakamoto coefficient indicates the minimum number of entities that can subvert the system. In the voting context, the entities can accept or reject the proposal in accordance with their interests, even if it may be against most users. Therefore, if power is significantly biased towards only a few voters, the DAO cannot be considered to attain true democracy, which is one of the main goals of DAOs. We calculated the Nakamoto coefficient based on the token weight of voters. We also ran a series of linear regression analyses, and logistic regression regressing the level of decentralization outcome of each DAO on the respective set of characteristics and controls, including secondary market value, market cap, DAO types, treasury, and voting power allocated for different proposals.

3.4 Limitation & Remediation

We have several inherent limitations to the data collection. The mutability of off-chain data, such as Snapshot proposals, is a significant limitation of our work. Our study utilized a crawler that accessed publicly available

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GraphQL APIs to collect voting data from platforms like Snapshot. However, this applied for proposals that edited ir deleted before voting period starts which we did not crawl in our dataset. According to official Snapshot documentation, it is possible to edit a proposal only before its voting period begins. Once a vote has started, the proposal becomes immutable, and changes can no longer be made, therefore, there is no risk of the edits in proposals that we collected. In cases where the proposal is canceled on Snapshot, the votes are still available, and thus the crawler was able to collect the data. This ensure the longitudinal validity of the result of Gini.

Second, DAO operations often involve nuanced cultural, social, and legal practices that cannot be fully captured by on-chain and publicly available datasets, particularly where informal practices or off-chain governance play a major role. Rather than claiming definitive conclusions, we frame our work as establishing empirical baselines and trends, while encouraging further mixed-method research to ground these in lived realities.

Third, token-based power definitions can obscure some subtleties of governance (e.g., delegation, abstention, inactive whales). However, our framework does not assume "1 token, 1 vote" as the foundation. Rather, it evaluates decentralization levels using effective voting power as implemented in practice across DAOs, including token-weighted, quadratic, and other systems. By systematically quantifying these patterns, our study contributes empirical evidence on how diverse governance mechanisms shape power distribution. In cases where DAOs employed non-linear mechanisms (e.g., quadratic voting, vote-escrow/veToken models), we used the effective voting weights as recorded by governance platforms, which already account for such mechanisms. These effective weights were then used as inputs for our decentralization metrics.

Our study qualitatively distinguishes between signaling and binding uses of Snapshot, however, our large-scale analysis does not model the full end-to-end governance pipeline for every DAO (e.g., ranging from proposal to Snapshot signal, on-chain execution, and finally treasury transfer). DAOs where Snapshot functions as a signaling mechanism, we do not claim that Snapshot votes directly reflect executed power rather used to characterize participation patterns and preference formation. We recommend the future by conducting systematic concordance studies between Snapshot signals and final on-chain outcomes across a broader set of DAOs to quantify alignment rates.

We based our analysis on explicit correlational and do not establish causal direction. For instance, we observed a relationship between market value and centralization, which may plausibly operate in both directions. Higher market capitalization may attract concentrated capital and exacerbate power inequalities, but it is also possible that pre-existing centralization, such as early insider allocation. Our results should be interpreted as descriptive of large-scale patterns rather than evidence of causal mechanisms.

Our analysis relies on data collected during 2019–2023, a period that substantially overlaps with a major cryptocurrency bull market. This market context may have amplified patterns of centralization by diluting the influence of early grassroots participants through speculative capital inflows. Nevertheless, this represents an important empirical contribution: similar market expansions are likely to recur, and our results provide a baseline for understanding how DAO governance evolves under high-growth, capital-influx scenarios.

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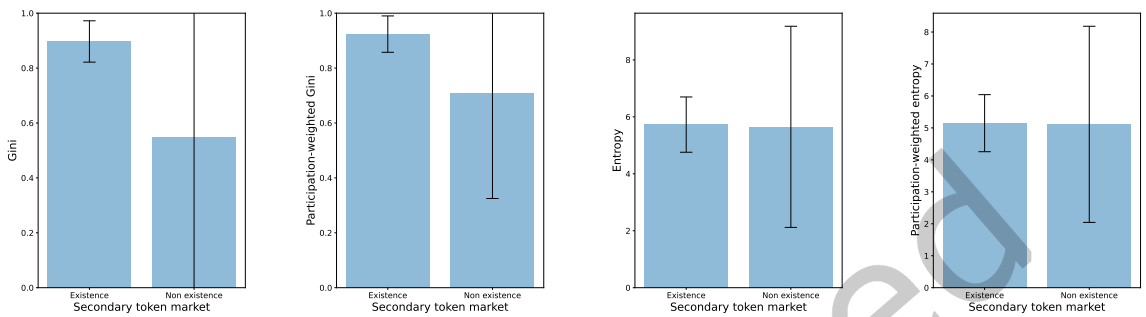
4 Result

4.1 Summary of 100 DAOs

Out of 100 DAOs analyzed, most used Snapshot for voting. Aavegotchi employed both Snapshot and Aragon smart contracts, while Uniswap used Snapshot for early voting signals and finalized votes on-chain. A few exceptions, such as RnDAO and LaborDAO, did not use formal voting platforms, opting instead for general polling and community discussions on platforms like Discord. Regarding governance tokens, 15% had no secondary market value, 67% had market value, and the rest were unspecified. In terms of treasury size, 42% of DAOs had treasuries between $0–$1M, 21% between $1M–$10M, 17% between $10M–$100M, 14% above $100M, and the rest were unspecified.



(a) Gini coefficient **(b)** Participation-weighted Gini **(c)** Entropy **(d)** Participation-weighted entropy

Fig. 1. Decentralization vs. secondary market value. Relationship between secondary market value and four decentralization metrics: (a) Gini coefficient, (b) participation-weighted Gini, (c) entropy, and (d) participation-weighted entropy.

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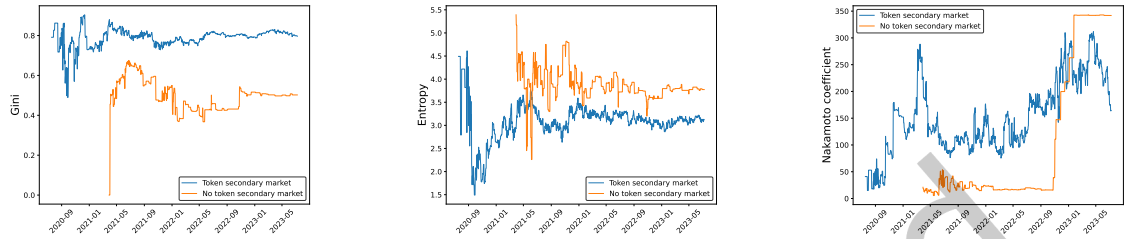
4.2 Different DAO Properties Impacting Level of Decentralization

Governance Token with the Secondary Market Value Associated with a Lower Level of Decentralization within a DAO. We run a series of t-tests to statistically compare a decentralization level between two groups: 1) DAOs with a secondary token market and 2) DAOs without a secondary token market. We measure the decentralization level using the three metrics: entropy, Gini, and Nakamoto coefficient. We find that both the average Gini index over time and the current Gini index have a significantly higher value in the first group consisting of DAOs with the secondary token market, indicating a lower degree of decentralization in terms of voting power distribution (average Gini index: $t \approx 5.049$, p-value< 0.001; current Gini index: $t \approx 3.469$, p-value< 0.001), in Figure 1. Similarly, the result of the average Gini with DAO users' participation rate weighted also indicates a small p-value (p-value$= 0.0002$, $t \approx 3.89$).Therefore, DAOs with a secondary token market tend to have a biased voting power distribution.

We then present the two groups' historical decentralization levels over time, measured by the three metrics, in Figure 2. In the graph, a data point presents average values across each group at some time point. Unlike the Gini index, the graphs show entropy between the two groups has been similar. Moreover, the relationship between the Nakamoto values of the two groups is not consistent. Before some point in time, the Nakamoto value in the DAOs without a secondary market was lower than the DAOs with a secondary market.

We also conduct an OLS regression with multiple variables, including the DAO type. Similar to the t-test, the result shows that whether to have a secondary market is a significant factor in determining their average decentralization level in terms of the Gini index (coeff = 0.3019, p-value = 0.807). On the contrary, it is shown that whether to have a secondary market is insignificant to determine their current Gini index when considering other factors such as their DAO type (coeff = 0.0584, p-value < 0.001). As a result, our analysis shows the existence of their secondary market is significantly associated with their voting power distribution.

Different DAO types are correlated with the level of decentralization. We employ ANOVA to statistically compare the levels of decentralization among multiple groups, each representing a distinct category of DAOs. Similar to the analyses above, we use the following three metrics to assess a decentralization level: entropy,

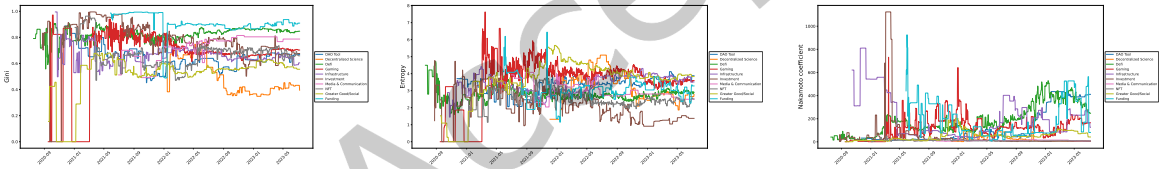


(a) Gini index vs DAO secondary market value

(b) Entropy vs DAO secondary market value

(c) Nakamoto coefficient vs DAO secondary market value

Fig. 2. Decentralization dynamics over time. Evolution of decentralization metrics for DAOs with secondary market value: (a) Gini index, (b) entropy, (c) Nakamoto coefficient.



(a) Average Gini coefficient vs DAO type

(b) Average entropy vs DAO type

(c) Average Nakamoto coefficient vs DAO type

Fig. 3. Decentralization dynamics by DAO type. Evolution of decentralization metrics across different DAO types: (a) Gini coefficient, (b) entropy, (c) Nakamoto coefficient.

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the Gini index, and the Nakamoto coefficient. We observed differences in decentralization levels across these various categories. Figure 3 presents a historical decentralization level for various DAO types. It shows that research, data analysis, social/greater good, and sports-oriented DAOs exhibit lower Gini values, indicating higher decentralization. In contrast, infrastructure, funding, product, and investment-oriented DAOs display higher Gini values, suggesting a lower degree of decentralization. Figure 7 in the Appendix also presents the same results. Relatedly, our analysis of the ANOVA test yields a Gini coefficient with a p-value of 0.05.

Moreover, when we run an OLS analysis with multiple variables, the social/greater good DAO type is a significant factor in determining the Gini index (coeff= 1.1545, p-value = 0.028). On the other hand, the DeFi and investment DAO types are significant factors regarding the Gini index (DeFi: coeff= 0.1121, p-value = 0.003; Investment: coeff= 0.1273, p-value = 0.070). As a result, we can conclude that the category of DAOs has an impact on decentralization levels, suggesting potential differences in entropy and Gini coefficient across different types of DAOs. It is important to note that various DAOs may belong to multiple categories, such as Radicle, API3, Osmosis, and Unlock, which may fall into both the DAO tools and Infrastructure categories, adding complexity to the analysis.

4.3 Evaluation of Decentralization Levels with Voting Power

In this section, "average voting power" refers to the mean effective voting weight of participants across all proposals within a DAO during the study period. This is distinct from confirmatory analyses in Section 4.2, which rely on the distribution of voting weights across all participants for hypothesis testing.

Pearson Correlation. We conduct a correlation analysis between the voting participation rate of a user and its voting power. If it shows a significant positive correlation, it can be a negative sign of a decentralization level cause more powerful voters tend to participate in voting more actively, underrepresenting minor voters more. In the analysis, we calculate a user's average voting power and the number of proposals that the user participated in voting. Then calculate a Pearson correlation coefficient. Table 1 summarizes a Pearson correlation coefficient and p-value of 100 DAOs. While the results were mixed, DAOs like 1inch, save, arbitrum, cakevote, club, fei, freerossdap, gmx, klima, gnosis, humanDAO, kcc, olympus, pie, sushiswap, etc - showed a significant positive correlation. This means that in those DAOs more powerful voters tend to participate in voting more actively, leading to a poor level of decentralization. Consequently, it appears that in many DAOs, wealthy voters are more likely to participate, exacerbating poor decentralization. LeagueDAO, partyDAO, peopleDAO, and treasure gaming, on the other hand, displayed a significantly strong negative correlation where participation from whale voters is not skewed.

Decentralization Level (Gini, Entropy, Nakamoto). Next, we analyze a decentralization level of voting mechanisms using three metrics: entropy, Gini, and Nakamoto. Before applying the three metrics, we first define the "recognized power" of a user as how much voting power a user has actually exercised. For example, even if a user has a lot of tokens that can be interpreted as voting power if the user rarely participates in voting in general, some may think they are not influential entities in a voting mechanism. Given the rationale, we calculate the "recognized power" as $V * P$, where V is a user's voting power (or token amounts) and P is the user's participation ratio.

We apply entropy and Gini to both voting power and the recognized power of users. On the other hand, Nakamoto is applied to only voting power because Nakamoto is to estimate the minimum number of users who can threaten a voting mechanism under the assumption that the users are corrupted. Entropy is calculated as $-\sum_i p_i log2(p_i)$, where p_i is a voting power (or recognized power) of user i / the total voting powers (or recognized powers) of all users. Gini is calculated as $\frac{\sum_i \sum_j |p_i - p_j|}{2 \sum_i \sum_j p_j}$. For Nakamoto, $\min_n(\sum_{i=1}^{n} q_i) >= Q$, where q_i is user i's voting power and Q is a quorum threshold, the minimum amount of voting power for a proposal to pass.

From Table 1, we can see many DAOs suffer from poor decentralization. In particular, 1inch, save, balancer, dydx, fei, gmx, gnosis, julswap, kogecoin. klime, lido, etc showed significantly poor decentralization, where their Gini coefficients were above 0.9 due to an extremely large power inequality or (and) the Entropy values were low due to a low number of participating voters. On the other hand, tosinshada, metricsdao, lexdao, olympus, goldhunt, golflinks, giaidao, graph protocol, daocity, friends with benefit showed the highest decentralization level among our target DAOs.

Decentralization Dynamics Overtime. We also examine the evolution of decentralization dynamics over time. In the Appendix, Table 1 provides a summary of linear regression results for 100 DAOs, highlighting significantly negative and positive trends in decentralization with orange and blue, respectively. Overall, a notable trend emerges, with a majority of DAOs in the Defi, Investment, and Infrastructure categories showing a consistently negative decentralization trend—indicating a decrease in decentralization levels over time. Figure 4 zooms in on Defi DAOs, for instance, SushiSwap, Lido, Ribbon, and Olympus DAO, revealing a significant increase in the Gini coefficient over time. This increase signifies a reduction in decentralization due to growing power inequality within these DAOs. It is worth noting that carbonswap, a decentralized exchange built on the security of Uniswap V2 contracts, offers fast and cost-effective DeFi services while also introducing new environmentally sustainable specifications.

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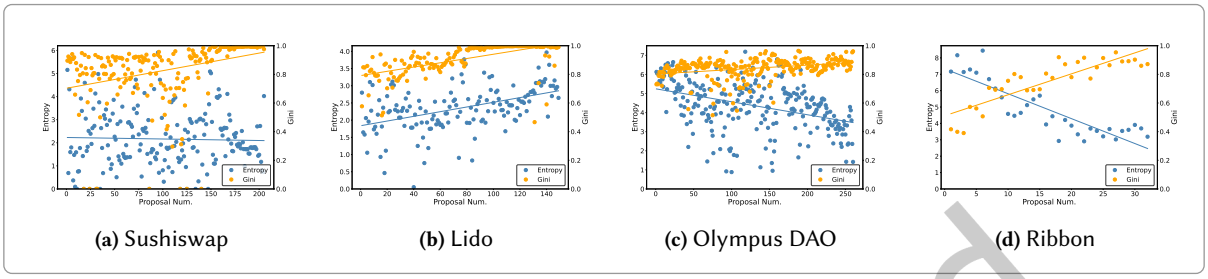
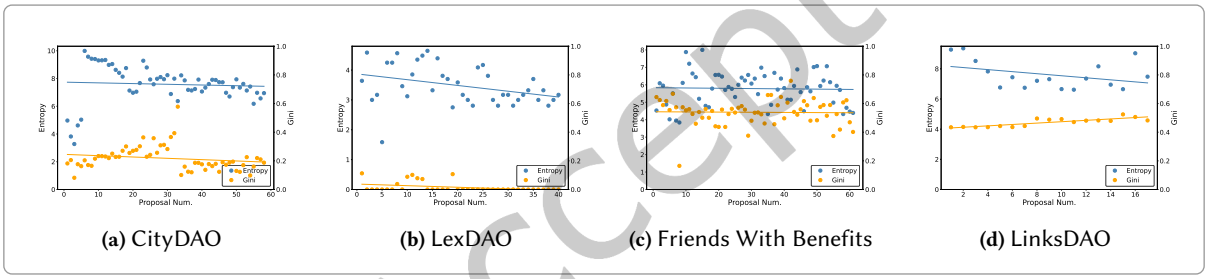


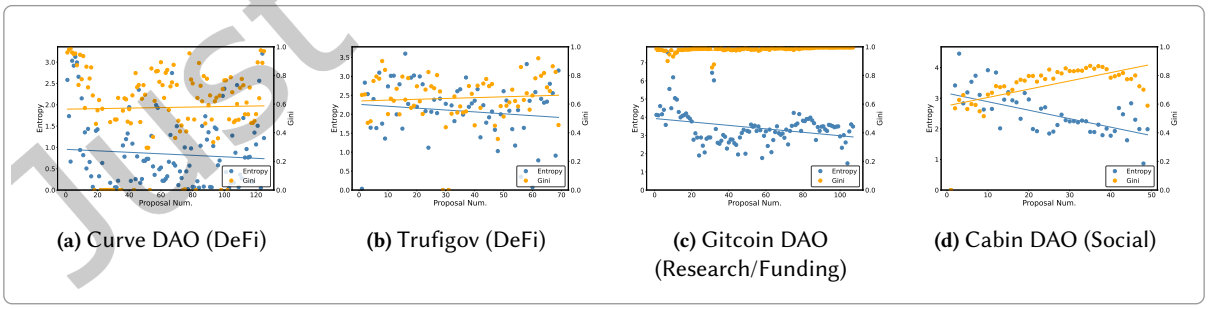
(a) Sushiswap (b) Lido (c) Olympus DAO (d) Ribbon

Fig. 4. Decentralization levels in example DeFi DAOs. Demonstrates increases and decreases in decentralization over time for representative DeFi DAOs.



(a) CityDAO (b) LexDAO (c) Friends With Benefits (d) LinksDAO

Fig. 5. Decentralization levels in example Social DAOs. Demonstrates increases and decreases in decentralization over time for representative Social DAOs.



(a) Curve DAO (DeFi) (b) Trufigov (DeFi) (c) Gitcoin DAO (Research/Funding) (d) Cabin DAO (Social)

Fig. 6. Decentralization exceptions across DAO types. Examples of DAOs where decentralization increases or decreases over time regardless of DAO type.

On the contrary, certain DAO types such as social and research, exemplified by CityDAO, LexDAO, and Friends with Benefits, exhibit a notably positive decentralization trend, signifying an increase in decentralization over time (Figure 5). It is worth noting that, as indicated in Table 1, in the CityDAO system, the issue of biased voter participation rates appears to be more pertinent than biased token weight distribution in terms of decentralization.

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The entropy is high, while the Gini coefficient for voters' token weight is low. Therefore, our primary focus for analyzing decentralization in CityDAO involves the linear regression results for entropy, as opposed to the Gini coefficient presented in Table 1. A qualitative observation is that CityDAO, for instance, employs quadratic voting, which may have contributed to the improvement in decentralization levels over time. Our findings also show that there are some exceptions of the decentralization level where there is an increase/decrease of decentralization regardless of the DAO type, such as curvedao – Defi Type, Trufigov – Defi Type, which has a slight increase trend of decentralization over time (Figure 6). Gitcoin DAO research or funding type and Canbin DAO Social DAO type have a decreasing trend of decentralization over time.

5 Discussion

In this section, we discuss the prevailing governance structure of DAOs and the degree to which they realize their envisioned goals as well as *open questions* of DAOs warranting further investigation. Based on the current outcomes of DAOs, we also provide *governance and design implications* for emerging technology governance.

5.1 DAOs as Complex Socio-Technical Systems

DAOs represent a potential empirical framework for complex social computing systems and an opportunity for understanding emerging forms of computational governance [17]. Through a computational lens, DAOs allow for modeling complex social, organizational, and legal issues as computational challenges [31, 76], offering new research directions. Prior literature has suggested factors such as token distribution, voter participation, delegation, and voting behavior patterns as influential in DAO governance success. Our findings confirm these hypotheses across a diverse set of DAOs, showing statistically significant correlations between secondary market activities and voting power distribution, thus decentralization. Specifically, we observe that DAOs focused on social causes or public good (e.g., sports-oriented DAOs) tend to exhibit lower Gini coefficients, reflecting higher decentralization. In contrast, infrastructure, and investment-oriented DAOs demonstrate higher Gini values leading to centralization. Additionally, whale voters in many DAOs, particularly in DeFi and investment categories, are more likely to participate, leading to poor decentralization. Notably, we found a significant power imbalance in DAOs like 1inch, Balancer, dYdX, Fei, GMX, and Gnosis, where low voter participation and high centralization are evident with entropy metrics.

Overall, a trend emerges where most DAOs in the DeFi, investment, and infrastructure sectors show a negative trend in decentralization, with the exception of a few DeFi DAOs, such as Trufigov, which exhibit slight decentralization improvements over time. Conversely, research- and funding-oriented DAOs, such as Gitcoin DAO, and social DAOs like Canbin DAO, show a decrease in decentralization, possibly due to governance contracts and community structures that were beyond the scope of this study. This indicates future research into the specific smart contract governance and off-chain community influencing these trends.

Throughout this study, the results of the Gini index metrics were consistent across both average and historical data points, suggesting that Gini metrics could serve as a useful tool for DAO stakeholders to monitor decentralization in the context of voting power and the influence of governance token design. However, the entropy and Nakamoto metrics were less consistent. While entropy showed some reliability in measuring the impact of voting participation on decentralization, our assessment was limited to active voters, which leaves out the potential centralizing effect of voters with closely aligned interests across elections, often referred to as voting bloc entropy [2]. This could be further empirically evaluated as a potential metric for decentralization. Another key factor we did not account for was inactive voters, particularly large token holders (whales). It would be valuable to explore the extent to which these inactive voters could influence outcomes if they participated, highlighting the potential importance of the Nakamoto coefficient in measuring the activity of both inactive

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whales who might disrupt outcomes and inactive grassroots voters who may need encouragement to better represent majority preferences.

While this paper focuses on large-scale empirical patterns of decentralization derived from realized voting power, we do not explicitly model, vote-escrow (veToken) mechanisms, which have become increasingly influential in DeFi and protocol governance [68]. In this case, governance power is not derived solely from token holdings; rather, participants lock tokens for a chosen duration to receive non-transferable voting escrow tokens (veTokens). Our work treats all systems through the lens of effective voting weights as documented from governance platforms, thus may lack the institutional properties of veToken models, such as lock-time stratification, governance power derivatives, bribery markets [2].

5.2 Democratic Infrastructures for Governing Technology

Our study highlights key metrics for decentralization, including voting power, grassroots participation, and non-monetary governance tokens. These factors significantly influence the design of decentralized, thus democratic systems aimed at engaging diverse stakeholders in collective decision-making. Traditionally, technology and algorithmic governance have been viewed through institutional or legal frameworks, focusing on regulatory mechanisms like laws and institutions [16], such as research on Internet protocol governance, NIST risk assessment [45] often centers on technical standards bodies. Recent studies have explored ways to involve users in the design and governance of digital technologies, including machine learning algorithms [63, 66], but these initiatives have not yet achieved widespread adoption. Engineering efforts within DAOs can provide valuable improvements for governance approaches in emerging areas such as AI governance, where the research community is exploring the democratization of AI [30, 52].

DAOs represent a community-oriented approach to technology governance, typically limited to protocols and contracts on a blockchain. The challenge lies in extending DAO governance to technologies like telecommunications protocols, and AI models development and deployment which are not implemented in a Web3 context. There is potential for addressing sensitive areas such as content moderation in social media or policing through algorithmic oversight, allowing affected parties to participate to govern the technology within their values and norms. This raises the question of how DAOs or similar social technologies can effectively govern a broader range of technologies beyond their current scope. Understanding best practices for DAO governance can help re-imagine and potentially redesign current political processes [8]. Considering both DAO technological innovation and theoretical underpinning from political science [8], we can design methods and tooling to engage diverse groups of users in governing AI models to collectively make a decision to shape AI rules whereas the technological components can enable transparent and trustworthy public participation, and minority voice representation.

5.3 Anecdotes to Metrics: Implication of Empirical Approaches

In this research, by translating decentralization into quantitative measures like the Gini coefficient, entropy, and Nakamoto coefficient, the study presents a structured approach to assess power dynamics in DAOs across a broad and varied sample. Although qualitative descriptions of DAOs offer depth, they often focus on specific cases. Our approach enhances these by providing replicable, and comparable metrics that enable practitioners, policymakers, and researchers to identify trends and patterns over time. Significantly, this framework does not replace the experiences of the community but establishes an empirical benchmark against which various governance experiments may be assessed. This is especially important in a rapidly changing field where anecdotal evidence may eclipse systematic findings [2]. Furthermore, our classification criteria (DAO type, treasury size, governance token structure, and voting mechanisms) can be regarded as a practical adaptation of existing DAO categorization frameworks, such as those proposed by Peña-Calvin et al.[56]. By aligning our sampling strategy

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with these theoretical models, we ensure our dataset captures both theoretical classifications and empirical variety, thereby reinforcing the connection between analytical frameworks and actual DAO practices.

For those actively involved in DAOs, the findings emphasize how design decisions regarding governance tokens, voting systems, and participation incentives directly influence decentralization results. Professionals can utilize these results as diagnostic markers: An elevated Gini coefficient or declining entropy may serve as alerts for centralization, encouraging corrective measures such as introducing quadratic voting, reputation-based tokens, or delegation frameworks. Evidence indicating that social and public-good DAOs trend towards greater decentralization suggests that mission-driven approaches are significant and that harmonizing tokenomics with collective objectives could enhance broader participation. On the other hand, the concentration noted in DeFi and investment DAOs highlights the difficulty of reconciling financial motivations with democratic governance, prompting leaders to adopt strategies that mitigate whale dominance.

From a more broader viewpoint, this research directly addresses long-standing discussions in political and management science regarding power, fairness, and institutional design. In conventional democracies, power is distributed through one-person-one-vote systems, yet DAOs explore alternatives, such as weighted voting, quadratic voting, and token-based governance. Our empirical data provide a unique opportunity to observe how these designs operate at scale. This contributes to management science by illustrating how digital organizations can self-govern without centralized hierarchies and to political science by broadening theories of representation and legitimacy in algorithmic settings [17]. The key takeaway is that power remains the essential element in democratic decision-making, irrespective of whether the system is analog or algorithmic. What DAOs offer is a collection of specific, verifiable mechanisms for redistributing influence in innovative ways. By documenting how these mechanisms function in practice, this study not only offers evidence for practitioners aiming for fairer and more resilient governance models but also lays a foundation for scholars and interdisciplinary groups interested in how emerging technologies transform collective decision-making possibilities [39].

Furthermore, our empirical findings become more salient when interpreted in conjunction with the decentralization. Our results suggest participation and centralization are tightly coupled components of a single governance dynamic while prior work emphasizes that governance power in DAOs is not only about token ownership but about activation of that power through participation, delegation, and coordination structures [19, 29, 78]. These concentration patterns may help explain the rise in governance attacks and manipulation in DAOs. Recent studies show problems such as vote buying, bribery markets, delegate capture, and coordinated voting groups that can distort governance outcomes [18, 28, 74]. Our study shows that a small subset of high-power voters disproportionately dominates participation in many DAOs, which indicates structural conditions under which such attacks become cheaper and more effective. Prior work shows that staking rules, rewards, lock-ups, and delegation strongly shape who votes and whose voices last over time [41]. Our results suggest that DAOs that mainly reward capital, without equally supporting broad participation, tend to become more centralized, as seen in falling entropy and rising Gini coefficients among financially intensive DAOs. By contrast, DAOs that emphasize social identity, recognize contributions, or use non-transferable governance rights tend to maintain more stable participation.

6 Conclusion

Our study found nuanced differences in DAO types, particularly in how the nature of governance tokens (monetary vs. non-monetary) impacts decentralization levels; token-based governance tends to disincentivize grassroots holders in the decision-making. Our work highlights increased grassroots participation and lower variance in voting power within a DAO correlates with a higher level of decentralization. This finding is closely related to core topics in political science, platform governance such as the transfer and allocation of power and one's influence in the community decision-making and the emergence and consequences of different governance systems. We

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conclude with the advantage of studying the governance of DAOs where it can serve as an interesting testbed for exploring social choice experiments, potentially improving the current governance structures in emerging technology, like AI.

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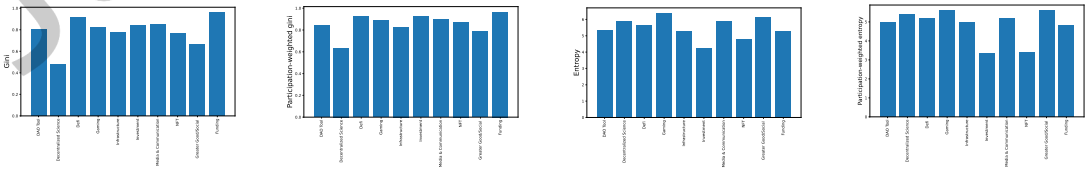
A Analysis of Average Decentralization Level Based on Different Factor



(a) average gini vs type **(b)** average participation gini vs type **(c)** average entropy vs type **(d)** average participation entropy vs type

Fig. 7. Average decentralization level (DAO Type) with Decentralization Metrics

Table 1. Evaluation of decentralization levels through correlation analysis and three metrics: Entropy, Gini coefficient, and Nakamoto.

Name	Corr (P-value)	Entropy	Entropy-P	Gini	Gini-P	Nakamoto
1inch.eth	0.014 (0.627)	5.189	4.272	0.974	0.985	8
aavegotch.eth	0.124 (0.000)	9.310	7.688	0.936	0.973	105
aave.eth	0.012 (0.003)	6.136	4.008	0.999	1.000	17
arbitrumfoundation.eth	0.003 (0.584)	5.653	5.741	0.985	0.986	5
balancer.eth	0.065 (0.000)	6.049	3.896	0.991	0.998	9
beetsdao.eth	-0.168 (0.149)	5.274	5.082	0.603	0.651	10
cabindao.eth	0.628 (0.000)	5.381	3.263	0.803	0.942	8
cakevote.eth	0.013 (0.000)	9.049	6.376	0.989	0.996	45
club.eth	0.046 (0.486)	5.573	5.477	0.830	0.848	10
council.graphprotocol.eth	NA	4.087	3.561	0	0.467	8
covalenthq.eth	-0.027 (0.638)	1.163	1.571	0.986	0.982	1
cream-finance.eth	0.086 (0.000)	4.394	3.574	0.994	0.996	4
curve.eth	-0.020 (0.610)	3.880	3.510	0.968	0.973	2
daocity.eth	-0.002 (0.902)	11.293	10.607	0.253	0.610	1046
daoofdiamonds.eth	0.159 (0.0430)	5.978	5.485	0.700	0.769	14
dappradar.eth	0.097 (0.002)	6.231	5.776	0.922	0.948	15
datacrunch.eth	0.386 (0.000)	5.567	4.880	0.733	0.826	11
devdao.eth	0.101 (0.000)	6.086	5.467	0.969	0.980	10
dydxgov.eth	-0.004 (0.780)	4.277	4.386	0.994	0.995	4
elfork.eth	0.769 (0.000)	5.364	5.075	0.412	0.562	18
fei.eth	0.056 (0.000)	6.844	5.775	0.973	0.989	19
ffdao.eth	0.292 (0.000)	5.658	4.471	0.829	0.919	9
freerossdao.eth	0.034 (0.357)	5.200	4.945	0.927	0.943	4
friendswithbenefits.eth	0.057 (0.006)	9.587	8.780	0.611	0.795	154
gaiadao.eth	0.060 (0.625)	4.856	4.706	0.685	0.715	7
gasdao.eth	0.085 (0.013)	6.530	5.610	0.886	0.927	15
gdao.eth	0.070 (0.441)	5.175	4.760	0.756	0.809	9
genomesdao.eth	-0.006 (0.967)	3.747	3.713	0.789	0.798	3
gitcoindao.eth	0.081 (0.000)	5.986	4.875	0.993	0.997	14
gmx.eth	0.055 (0.000)	7.453	6.987	0.996	0.997	29
gnosis.eth	0.010 (0.478)	5.549	4.049	0.991	0.996	7
goldhuntdao.eth	0.416 (0.000)	7.052	6.063	0.461	0.717	33
golflinks.eth	0.098 (0.000)	10.413	9.848	0.464	0.648	450
gov.radicle.eth	0.285 (0.000)	4.090	3.032	0.961	0.981	4
humandao.eth	0.024 (0.787)	3.106	2.967	0.923	0.930	2
instadapp-gov.eth	-0.023 (0.830)	2.995	3.217	0.935	0.925	2
insuredao.eth	0.162 (0.234)	3.437	3.233	0.848	0.870	3
julswap.eth	-0.007 (0.638)	3.353	3.465	0.995	0.995	2
karastar.eth	-0.070 (0.248)	4.896	4.263	0.908	0.918	7
kcc.eth	0.031 (0.158)	9.028	8.596	0.791	0.826	185
kleros.eth	0.055 (0.332)	5.617	5.527	0.865	0.882	9

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Table 1. Evaluation of decentralization levels through correlation analysis and three metrics: Entropy, Gini coefficient, and Nakamoto (continued).

Name	Corr (P-value)	Entropy	Entropy-P	Gini	Gini-P	Nakamoto
klimadao.eth	0.015 (0.183)	5.843	6.529	0.976	0.982	4
kogecoin.eth	0.103 (0.003)	0.286	0.187	0.999	0.999	1
kongsdao.eth	0.409 (0.000)	6.446	4.728	0.531	0.837	22
leaguedao.eth	-0.234 (0.092)	3.355	3.620	0.844	0.815	3
lexdao.eth	-0.003 (0.982)	6.277	5.493	0.021	0.564	39
lido-snapshot.eth	0.081 (0.000)	5.654	4.119	0.993	0.998	12
loopringdao.eth	0.055 (0.001)	8.986	8.518	0.849	0.882	88
loot-dao.eth	0.152 (0.000)	8.591	7.380	0.638	0.806	70
macaronswap.eth	0.165 (0.001)	6.899	6.283	0.764	0.826	23
matrixdaoresearch.eth	NA	6.129	5.981	0.000	0.240	35
meebitsdao.eth	0.299 (0.003)	3.036	2.406	0.876	0.947	2
nftx.eth	0.390 (0.000)	5.168	2.346	0.920	0.983	5
officialoceandao.eth	0.047 (0.203)	6.277	6.062	0.889	0.922	11
olympusdao.eth	0.006 (0.582)	7.624	7.799	0.957	0.966	21
opcollective.eth	0.033 (0.000)	6.285	5.240	0.991	0.996	10
partydao.eth	-0.153 (0.140)	3.024	3.667	0.933	0.895	3
people-dao.eth	-0.025 (0.191)	6.110	4.906	0.963	0.975	9
piedao.eth	0.015 (0.653)	5.102	6.017	0.946	0.939	6
pleasrdao.eth	0.429 (0.000)	5.054	3.800	0.710	0.864	9
polygonvalidators.eth	NA	5.459	5.459	0.000	0.000	22
polywrap.eth	0.845 (0.000)	4.121	2.211	0.763	0.931	4
pooltogether.eth	0.061 (0.059)	6.303	5.821	0.931	0.949	16
primexyz.eth	-0.196 (0.239)	1.490	2.278	0.938	0.898	1
raid-guild	0.048 (0.821)	3.560	3.430	0.620	0.668	3
rarible.eth	-0.006 (0.785)	4.554	5.121	0.975	0.977	2
rbn.eth	-0.010 (0.711)	7.714	7.790	0.802	0.829	25
rook.eth	0.066 (0.229)	6.455	5.912	0.798	0.856	20
shapeshiftdao.eth	0.030 (0.191)	6.068	5.796	0.963	0.976	12
sharkdao.eth	0.339 (0.000)	7.045	4.389	0.820	0.958	25
sismo.eth	0.064 (0.000)	9.801	8.771	0.758	0.869	129
snapshot.dcl.eth	0.033 (0.006)	6.812	5.651	0.979	0.993	15
songadao.eth	NA	5.907	5.534	0.000	0.389	30
stgdao.eth	0.008 (0.004)	4.823	4.697	0.989	0.995	4
sushigov.eth	0.013 (0.176)	5.716	5.139	0.994	0.996	9
tempusgov.eth	-0.067 (0.469)	4.356	4.662	0.853	0.832	4
temp.metricsdao.eth	-0.152 (0.376)	5.155	4.793	0.026	0.398	18
theplantdao.eth	0.207 (0.138)	5.016	4.922	0.511	0.551	12
tomoondao.eth	0.128 (0.000)	8.785	7.725	0.857	0.915	88
tosinshada.eth	NA	3.585	3.585	0.000	0.000	6
trashdao.szns.eth	0.468 (0.016)	3.342	2.247	0.691	0.832	3
treasuregaming.eth	-0.033 (0.037)	4.697	4.468	0.995	0.995	7

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Table 1. Evaluation of decentralization levels through correlation analysis and three metrics: Entropy, Gini coefficient, and Nakamoto (continued).

Name	Corr (P-value)	Entropy	Entropy-P	Gini	Gini-P	Nakamoto
truefigov.eth	0.300 (0.000)	4.642	3.860	0.901	0.943	5
trustwallet	0.074 (0.000)	8.576	8.108	0.950	0.963	56
ubi-voting.eth	0.285 (0.000)	3.058	2.514	0.940	0.972	2
uniswap	-0.004 (0.527)	5.509	4.890	0.999	0.999	13
unlock-protocol.eth	-0.028 (0.722)	2.623	3.845	0.948	0.928	1
vote-perp.eth	0.019 (0.512)	5.175	4.575	0.973	0.981	8
vote.vitadao.eth	0.052 (0.346)	4.654	4.770	0.929	0.936	5
yam.eth	0.144 (0.000)	6.825	5.175	0.979	0.991	19
yearn	0.064 (0.000)	6.714	6.873	0.961	0.964	15
Average	0.100	5.618	5.04	0.794	0.857	35.923
(Std)	(0.186)	(1.989)	(1.834)	(0.275)	(0.202)	(120.432)

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